

ECON 381: Practice Midterm 1

Professor MUYANG REN

Fall 2026

Name: _____

Instructions

1. Write your name on this exam, your note sheet, and every answer sheet.
2. **Do not open this exam until instructed to do so.**
3. Silence your cell phone and place it in your bag, not in your pocket.
4. This exam is worth 100 points, with 5 additional bonus points available. The maximum recorded score is 100.
5. Check that you have all 7 pages. You have 75 minutes to complete the exam.
6. You may use one 8.5×11 sheet of handwritten notes, on both sides, prepared before the exam. Turn in your note sheet with your exam and answer sheets.
7. Write all answers and supporting work on separate blank sheets of paper. Do not write answers on the exam pages. Clearly number each answer page, label each answer with its section, question number, and part. For each multiple-choice question, write the question number and the corresponding one answer letter.
8. Use other blank sheets for scratch work. Clearly cross out any work you do not want graded in your answer sheets.
9. One standalone, non-programmable calculator is allowed. All other electronic devices are prohibited. For confidence intervals, round the standard normal critical values discussed in class to two decimal places.
10. Sign the honor pledge below. At the end of the exam, turn in this exam, your answer sheets, and your note sheet.
11. **Good luck!**

I pledge on my honor as a University of Tennessee student that I will neither give nor receive aid on this examination.

Signature: _____

1. Multiple Choice Questions (30 points total; 10 questions; 3 points each)

Instructions: On your separate answer sheets, write each question number and **one** answer letter (a, b, c, or d). No explanation is required. Selecting more than one answer earns no credit for that question.

1. In the California school dataset, one observation represents
 - (a) one student.
 - (b) one teacher.
 - (c) one school district.
 - (d) one random sample of 420 districts.

2. The fitted line is $\widehat{\text{TestScore}} = 698.9 - 2.28 \text{STR}$. If the student–teacher ratio decreases by 2, the fitted test score
 - (a) decreases by 4.56 points.
 - (b) increases by 4.56 points.
 - (c) increases by 2.28 points.
 - (d) decreases by 2.28 points.

3. In $\text{TestScore}_i = \beta_0 + \beta_1 \text{STR}_i + u_i$, the population regression error u_i represents
 - (a) the estimated slope of the sample regression line.
 - (b) the student–teacher ratio in district i .
 - (c) the fitted test score from the estimated sample line.
 - (d) other influences on test scores not represented by STR_i in the population line.

4. Which statement correctly distinguishes the population mean μ_Y from the sample average $\bar{Y}$?
 - (a) μ_Y is fixed but typically unknown; $\bar{Y}$ generally varies across random samples.
 - (b) μ_Y changes whenever a new sample is drawn; $\bar{Y}$ is fixed across samples.
 - (c) Both are known before collecting any data.
 - (d) Every random sample has $\bar{Y} = \mu_Y$.

5. What does $E[\text{TestScore} \mid C = \text{Los Angeles}]$ mean, where C denotes county?
 - (a) The population average test score across all counties combined.
 - (b) The sample average test score among districts in Los Angeles County.
 - (c) The population average test score among districts in Los Angeles County.
 - (d) The probability that a district is in Los Angeles County.

6. Income is converted from thousands of dollars to dollars, while test scores remain in the same units. What happens to the correlation between income and test scores?
- (a) It is multiplied by 1,000.
 - (b) It stays the same.
 - (c) It is divided by 1,000.
 - (d) Its sign reverses.
7. Using the same sample and the Central Limit Theorem, a 95% confidence interval for the population mean is
- (a) narrower than the 90% interval and has a different center.
 - (b) the same width as the 90% interval.
 - (c) wider than the 90% interval and has a different center.
 - (d) wider than the 90% interval and has the same center.
8. Ordinary least squares chooses the intercept and slope to minimize
- (a) $\sum_{i=1}^n [Y_i - (b_0 + b_1 X_i)]^2$.
 - (b) $\sum_{i=1}^n [Y_i - (b_0 + b_1 X_i)]$.
 - (c) $\sum_{i=1}^n (X_i - \bar{X})^2$.
 - (d) $\sum_{i=1}^n (Y_i - \bar{Y})^2$.
9. In an OLS regression with an intercept and variation in X , the fitted line must pass through
- (a) $(0, 0)$.
 - (b) every observed data point.
 - (c) $(\bar{X}, \bar{Y})$.
 - (d) $(\bar{X}, 0)$.
10. The standard error of the regression (SER) describes
- (a) the fraction of sample variation in Y captured by the fitted line.
 - (b) the typical residual size, measured in the units of Y .
 - (c) the standard error of the sample mean $\bar{Y}$.
 - (d) the estimated slope of the fitted line.

2. True or False Questions (20 points total; 4 questions; 5 points each)

Instructions: On your separate answer sheets, label each part and write **T** or **F**. You must justify each answer. Full credit requires a correct answer and justification.

- (a) The Law of Large Numbers guarantees that adding one more observation to a random sample always brings its sample mean closer to the population mean.

- (b) The sample standard deviation s_Y describes the spread of individual outcomes, while $SE(\bar{Y})$ estimates the spread of sample averages across repeated random samples of the same size.

- (c) A 95% confidence interval for the population mean is designed to contain 95% of the individual outcomes in the population.

- (d) If X is multiplied by 2 while Y is unchanged, then $\text{Cov}(2X, Y) = 2\text{Cov}(X, Y)$, provided the covariances exist.

3. Analytical Question (50 points total)

Instructions: Write your answers and show all work on your separate answer sheets. Label each part (a)–(i). Use the same three observations throughout. Parts (a)–(e) concern the population mean of Y ; parts (f)–(i) concern the regression of Y on X with an intercept.

Consider this hypothetical random sample of three observations:

Observation i	X_i	Y_i
1	1	3
2	2	3
3	3	6

- (a) (4 points; 2 each) Compute the sample averages $\bar{X}$ and $\bar{Y}$.
- (b) (5 points) Compute the sample standard deviation s_Y , using the sample variance with divisor $n - 1$. You may leave a square root in your answer.

$$\text{Hint: } s_Y = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2}.$$

- (c) (4 points) Compute the standard error of the sample average, $SE(\bar{Y})$.
- (d) (8 points; 4 each) Compute the 90% and 95% confidence intervals for the population mean μ_Y . Round the critical values to two decimal places and report endpoints to two decimal places.
- (e) (4 points) With only these three observations, does the procedure used to construct the 95% confidence interval in part (d) necessarily cover the population mean in 95% of repeated samples? Explain.

Analytical Question (continued)

The data remain $(X_i, Y_i) = (1, 3), (2, 3), (3, 6)$. Estimate

$$Y_i = \beta_0 + \beta_1 X_i + u_i.$$

(f) (10 points) Compute the OLS slope estimator $\hat{\beta}_1$. Show the numerator and denominator of the slope formula using deviations from the sample means.

(g) (6 points) Compute the OLS intercept estimator $\hat{\beta}_0$ and write the fitted regression equation.

(h) (6 points; 1 per entry) Compute the three fitted values $\hat{Y}_i$ and three residuals $\hat{u}_i$. Copy the table onto your separate answer sheets and fill in the fitted values and residuals there.

i	X_i	Y_i	$\hat{Y}_i$	$\hat{u}_i$
1	1	3		
2	2	3		
3	3	6		

(i) (3 points) Compute R^2 . Show the total sum of squares (TSS) and the residual sum of squares (RSS) used in your calculation.

4. Bonus Question (5 points)

Instructions: Write your answers on your separate answer sheets and label parts (a) and (b). Explain your reasoning using the OLS formulas or minimization problem discussed in class.

A researcher observes three outcomes, $Y_1 = 3$, $Y_2 = 3$, and $Y_3 = 6$, but now the regressor has the same value in every observation:

$$X_1 = X_2 = X_3 = 2.$$

Observation i	X_i	Y_i
1	2	3
2	2	3
3	2	6

The researcher wants to fit $Y_i = \beta_0 + \beta_1 X_i + u_i$ with both an intercept and a slope.

- (a) (3 points) Can the usual OLS formulas produce a **well-defined** slope? Explain what happens to the numerator and denominator of the slope formula.

- (b) (2 points) If instead the intercept is fixed at a specified value b_0 , is there a unique slope minimizing the sum of squared residuals? Express that slope in terms of b_0 .